

ϑ : communication threshold

independent
of

q_r : server step magnitude

each client $i = 1, \dots, M$

server

all clients

1 LOCAL SGD

$$\mathbf{w}_{i,0}^r = \mathbf{w}^r$$

$$\boldsymbol{\delta}_i^r = -\eta_r \sum_{e=0}^{E_r-1} \mathbf{g}_{i,e}^r$$

local model change

2 LEAKY EVIDENCE

$$\tilde{\mathbf{z}}_i^r = \rho_r \mathbf{z}_i^r + \frac{p_i}{\eta_r} \boldsymbol{\delta}_i^r$$

state carried
across rounds

3 THRESHOLD AND RESET

$$m_{ij}^r = \mathbf{1}_{\{|\tilde{z}_{ij}^r| \geq \vartheta\}}$$

$$c_{ij}^r = m_{ij}^r \text{sign}(\tilde{z}_{ij}^r)$$

$$z_{ij}^{r+1} = (1 - m_{ij}^r) \tilde{z}_{ij}^r$$

if fired: send ± 1 ; reset
state to zero

sparse
uplink
address j
+ sign

4 SERVER AGGREGATION

$$\mathbf{C}^r = \sum_{i=1}^M \mathbf{c}_i^r$$

$$\mathbf{w}^{r+1} = \mathbf{w}^r + q_r \mathbf{C}^r$$

pulses reinforce or
cancel;
net pulse scaled by
 q_r

5 EXACT DOWNLINK EVERY ROUND

replay ordered sparse events
or
receive dense checkpoint

$$\mathbf{w}_{i,0}^{r+1} = \mathbf{w}^{r+1}, \quad i = 1, \dots, M$$

clients exactly match
the server

next round starts from $\mathbf{w}^{r+1}$